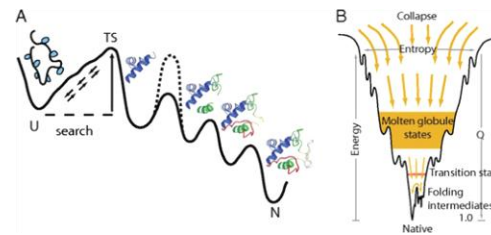


Bioinformatics

Protein Structure Prediction

Prof. David Jones
d.t.jones@ucl.ac.uk

The Protein Folding Problem

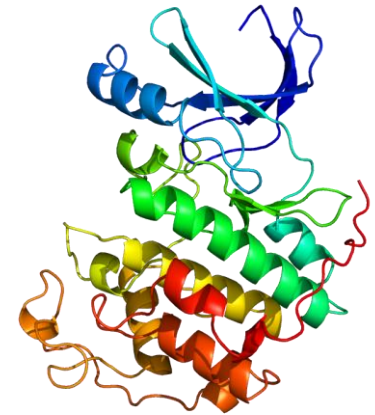


MENFQKVEKIGEGTYGVVYKARNKLTGEVVALKKIRLDTETEGVPSTAIREISLLKEL...



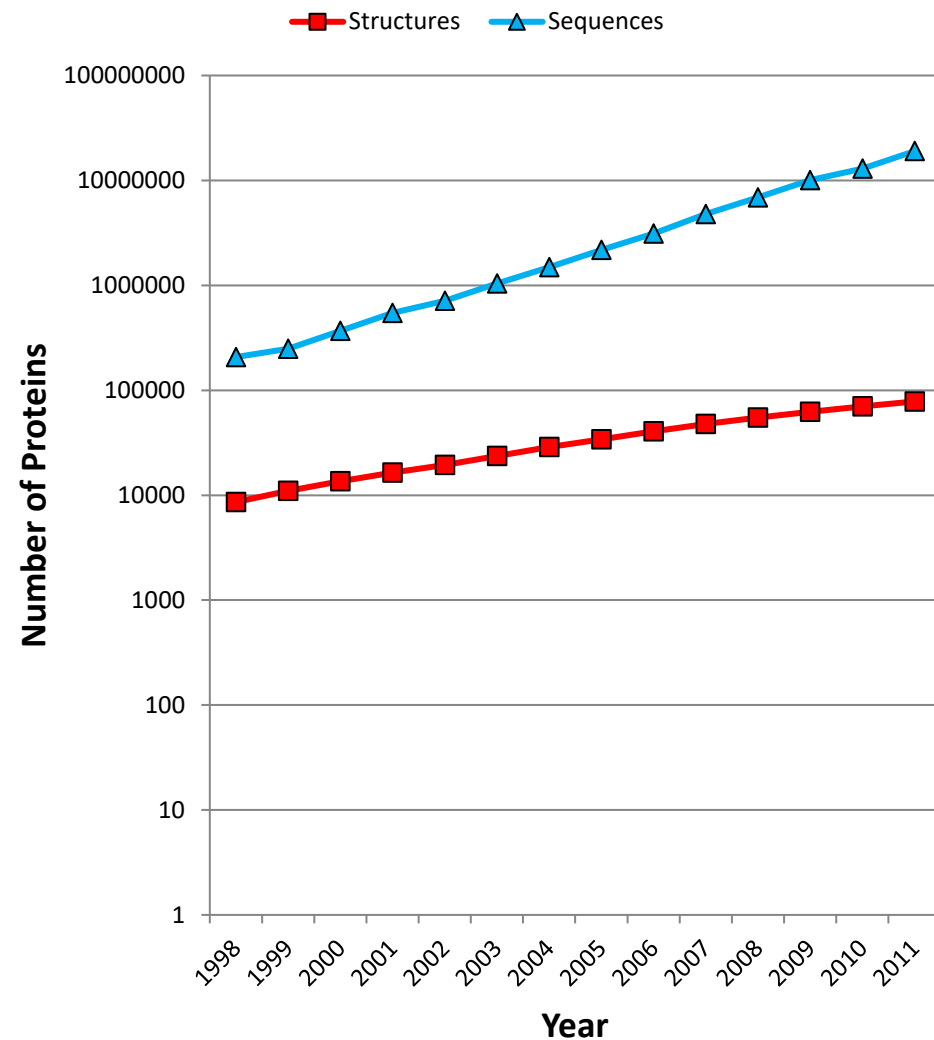
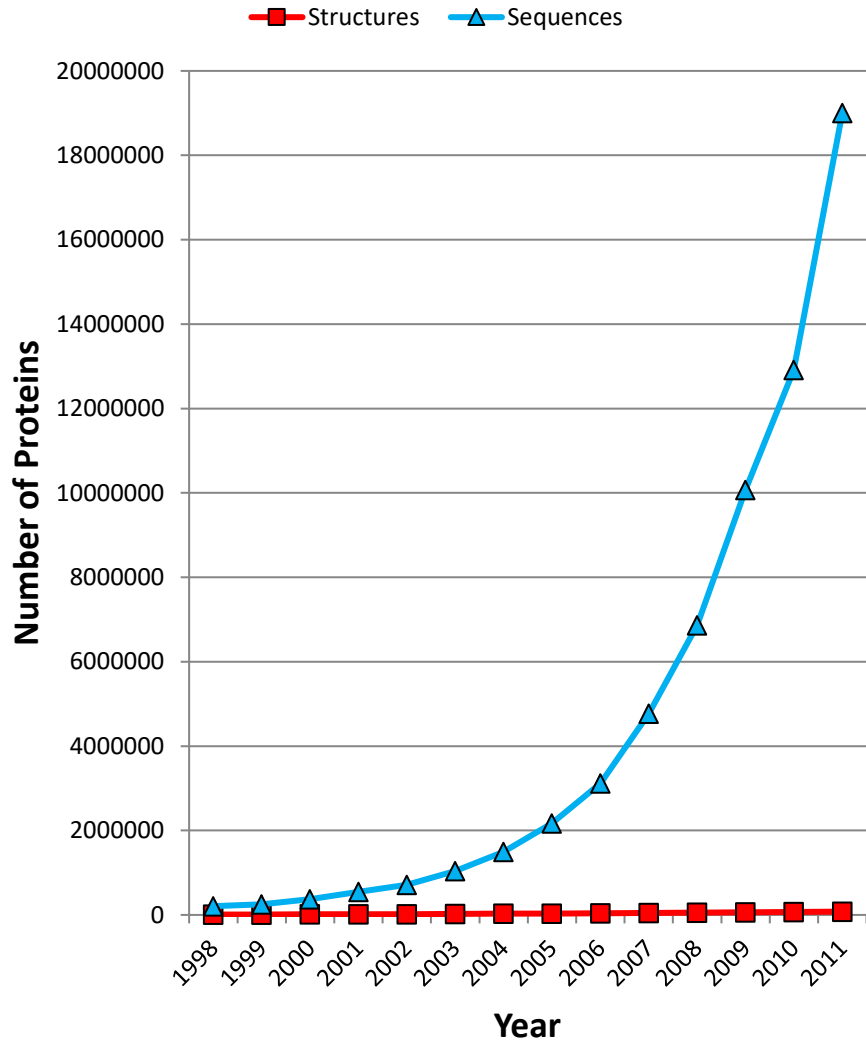
Genetically
encoded amino
acid sequence
(primary structure)

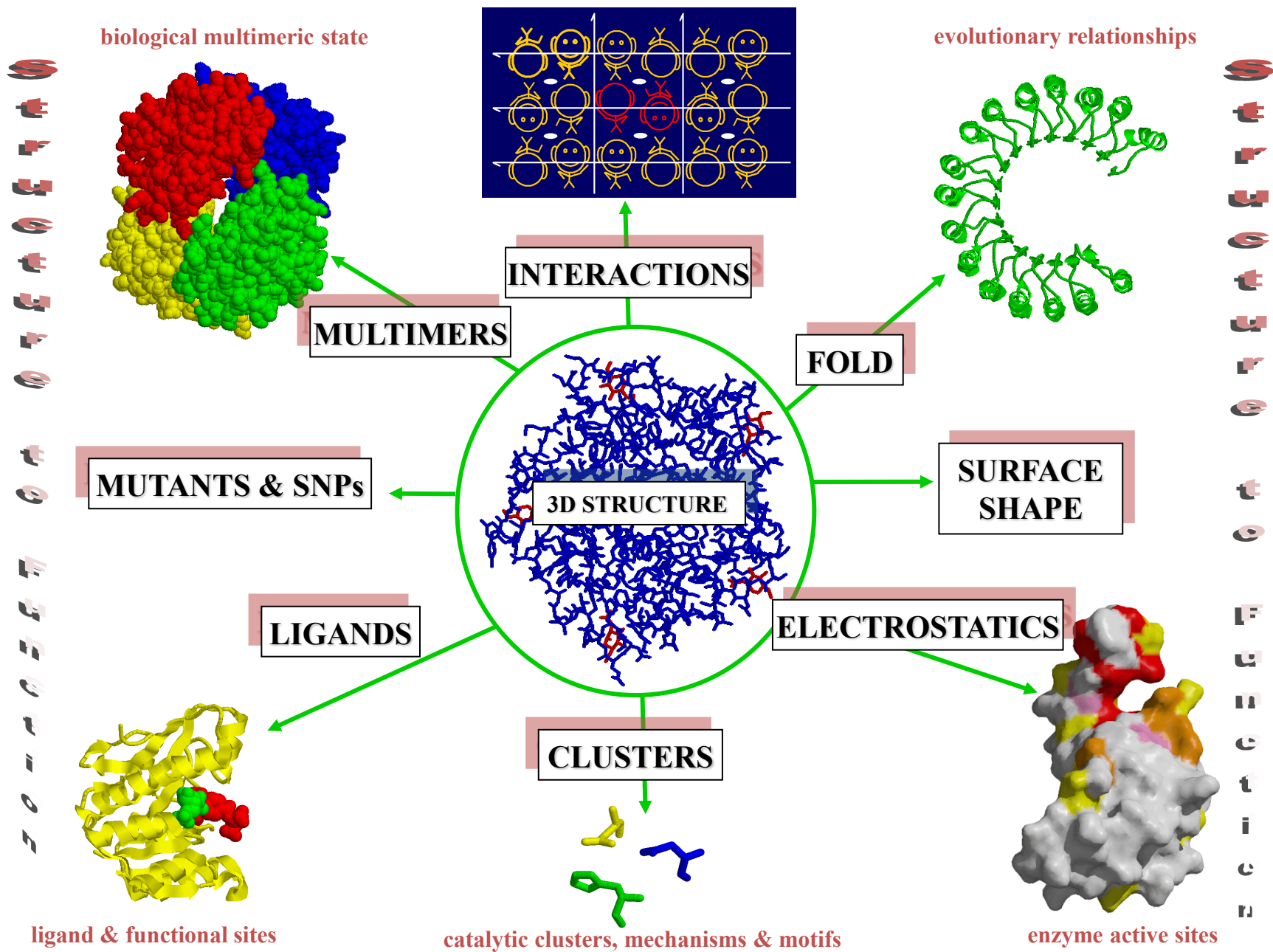
```
chn[0].c[2] = 0.0;
setdihedral(chn[0].n, chn[0].ca, chn[0].c, chn[0].a, 1.0, 0)
for (i=1; i<seqlen; i++)
{
  setdihedral(chn[i-1].n, chn[i-1].ca, chn[i-1].c, chn[i-1].a, 1.0, 0)
  setdihedral(chn[i-1].c, chn[i-1].ca, chn[i-1].c, chn[i-1].a, 1.0, 0)
  setdihedral(chn[i-1].c, chn[i-1].ca, chn[i-1].c, chn[i-1].a, 1.0, 0)
  setdihedral(chn[i-1].c, chn[i-1].ca, chn[i-1].c, chn[i-1].a, 1.0, 0)
}
```



Natively folded 3-D
structure
(tertiary structure)

Growth of sequence and structure data banks







Physics teaching
UK physicists bring
the school syllabus
up to date
p708



Laser target
The US Nova laser
target chamber goes
to France on loan
p709



Mars blow
The loss of the Mars
Polar Lander gives
NASA a headache
p711



X-ray success
European space
scientists relieved
at XMM launch
p713

IBM promises scientists 500-fold leap in supercomputing power...

Paris

Biologists are hailing IBM's US\$100 million

capable of more than one quadrillion floating point operations ('flops') per sec-

... and a chance to tackle protein structure

Paris

IBM's choice of protein folding as the focus of its project to develop the supercomputer Blue Gene is part of a strategy to position itself as a leader in computational biology, as the post-sequencing phase of the Human Genome Project approaches (see above).

The company has regrouped its life-science activities into a single unit focused on genomics and drug design. Sharon Nunes, senior research manager at IBM's Computational Biology Center at the company's Deep Computing Institute in New York, says that predicting protein structure will be a key use of genome data. "Using today's computers it would take a thousand years to fold a protein," she says.

"Blue Gene is tremendously

exciting," says Christopher Dobson, who works on protein folding at the University of Oxford. "Folding is how the cell converts information from DNA into biological function, how the cell targets proteins to particular parts of the cell, controls and localizes them, and is key to turning activity on and off," he says.

Some are sceptical. Computing power alone is not enough, says Geoff Barton, a scientist at the European Bioinformatics Institute in Cambridge. He argues that solving folding will require enormous progress in understanding the interactions between the atoms in a protein.

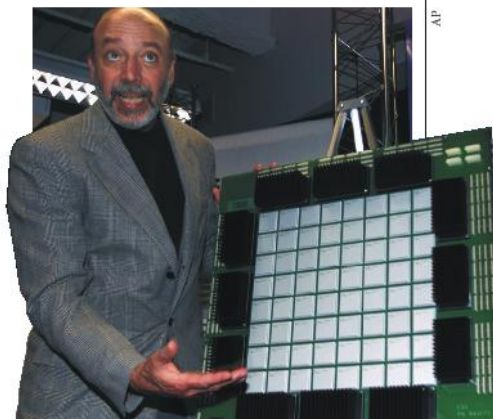
"There is no guarantee that if you find the minimum energy conformation it will actually be the

native protein conformation, because the potentials used to model the pairwise interactions are imperfect. There are also difficulties with modelling the true solvent environment of the protein."

Ajay Royyuru, a researcher at IBM's Computational Biology Center, says this will be a major thrust of the Blue Gene project. Dobson says the timing is right in that "many of the fundamental principles of folding are beginning to emerge".

Whatever the reservations about protein folding, few biologists doubt that, if it succeeds, the project's contribution to handling the mountains of data emerging from genome sequencing programmes will be revolutionary. **D.B.**

1 two million times more today's top desktops. Most predicted that fundamental difficulties would prevent a computer being built before the end of the century (see *Nature* 402 supp., C67). IBM is doing this," says a scientist at the University of Oxford and NASA project scientist. IBM is showing the performance computing in the new generation of supercomputers, he says. Technological constraints on the petaflop machine have been easing the speed with which the memory — and reducing



On the board: IBM senior vice-president Paul Horn with a mock-up of a Blue Gene array.

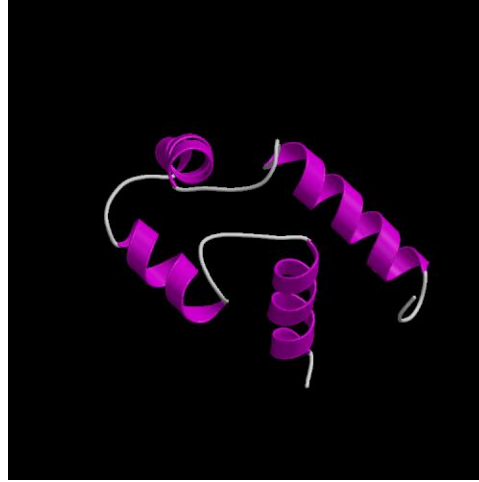
NATURE | VOL 402 | 16 DECEMBER 1999 | www.nature.com

CASP

- **CASP (Critical Assessment of protein Structure Prediction)** is a community-wide, worldwide experiment for protein structure prediction that has taken place every two years since 1994.
- CASP provides research groups with an opportunity to objectively test their structure prediction methods and delivers an independent assessment of the state-of-the-art.
- Insightful Wikipedia quote: “Even though the primary goal of CASP is to help advance the methods ... many view the experiment more as a ‘world championship’ in this field of science. More than 100 research groups from all over the world participate in CASP on a regular basis and it is not uncommon for entire groups to suspend their other research for months while they focus on getting their servers ready for the experiment and on performing the detailed predictions.”



Protein Secondary Structure



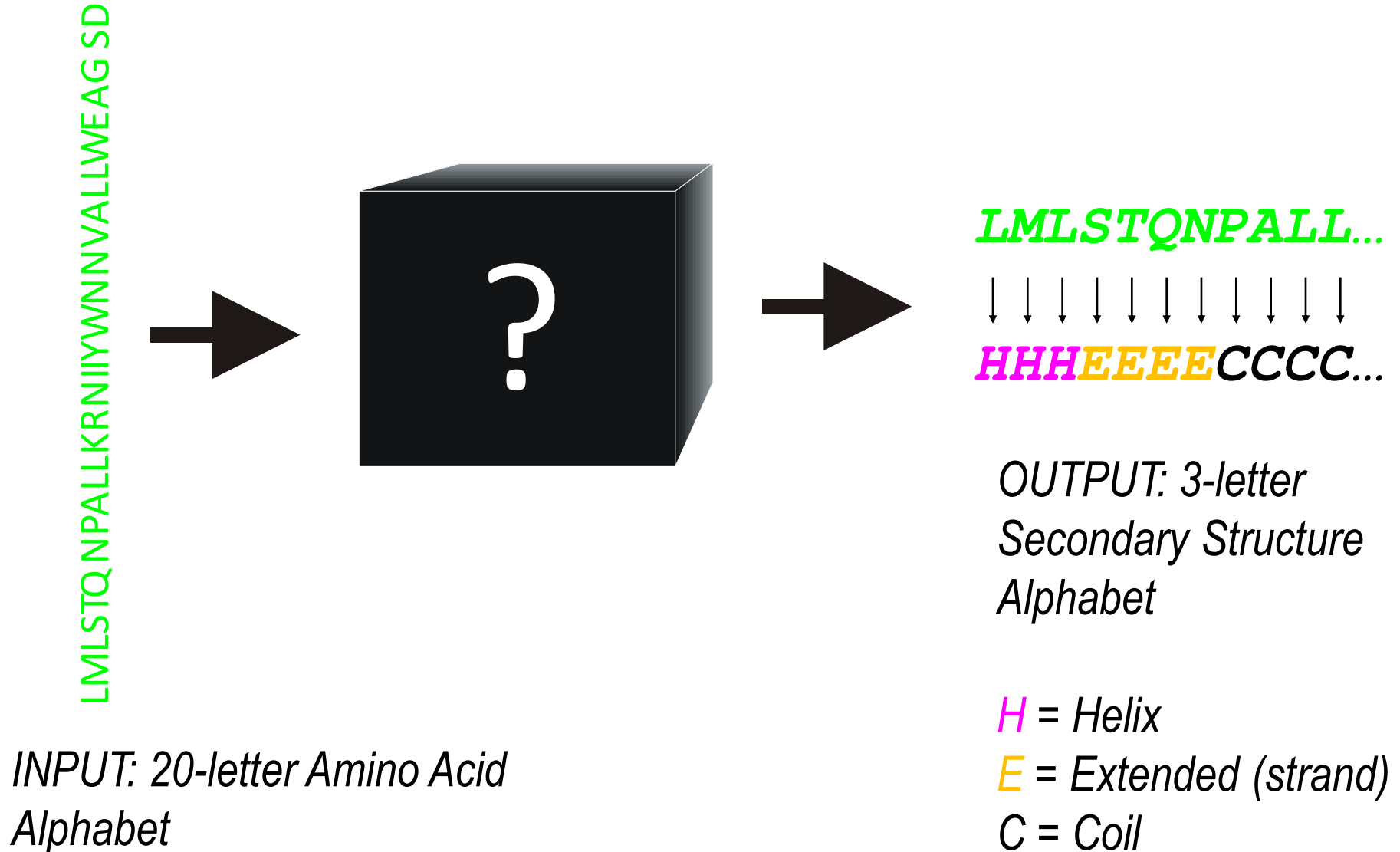
ALPHA HELIX

***BETA
SHEET/STRAND***



COIL/LOOP

Secondary Structure Prediction



1st Generation Methods

- Based on statistical analysis of single amino acid properties
- Examples:
 - Chou & Fasman (1974)
 - Lim (1974)
 - Garnier, Osguthorpe & Robson (1978)

2nd Generation Methods

- Based on peptide segments / residue pairs
- Examples:
 - GOR III (1987)
 - Directional tables / information theory
 - **Neural Networks**
 - **Qian & Sejnowski (1988)**
 - **Bohr et al. (1988)**
 - **Holley & Karplus (1989)**

Representation

- Usual representation is to use 21 inputs per amino acid with one-hot encoding:

Ala : 100000000000000000000000

Val : 000000000000000000000010

--- : 000000000000000000000001

Predicting Protein Secondary Structure by Machine Learning

Rather than just one residue at a time, we look at neighbouring residues as well!

*Window of
15 residues*

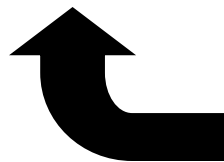
MLSPQAMSDFH**E**ELKWLLCNIPGQKLASLANREYT...



→ *Helix*

→ *Strand*

→ *Coil*



*We are predicting the secondary
structure for this central residue*

3rd Generation Neural Network Methods

PHD

Rost, B. & Sander, C. (1993) Prediction of protein secondary structure at better than 70% accuracy. J. Mol. Biol. 232, 584-599.

PSIPRED

Jones, D.T. (1999) Protein secondary structure prediction based on position-specific scoring matrices. J. Mol. Biol. 292: 195-202.

3rd Generation Methods

- Key idea was to replace the single-sequence one-hot encoding with log-likelihood ratio scores from profiles (profile-HMM match states)
- Allowed exploitation of evolutionary information
 - Can extract long-range information by relating CONSERVATION to SOLVENT ACCESSIBILITY
 - Conserved Hydrophobic Residues -> BURIED residue
 - Variable Polar Residues -> EXPOSED residue

Pros & Cons of 3rd Generation Methods

- PROs
 - High residue accuracy
 - Less underprediction of strands
 - Good quality segment predictions
- CONS
 - Provides prediction for FAMILY CONSENSUS structure NOT THE STRUCTURE OF THE TARGET SEQUENCE

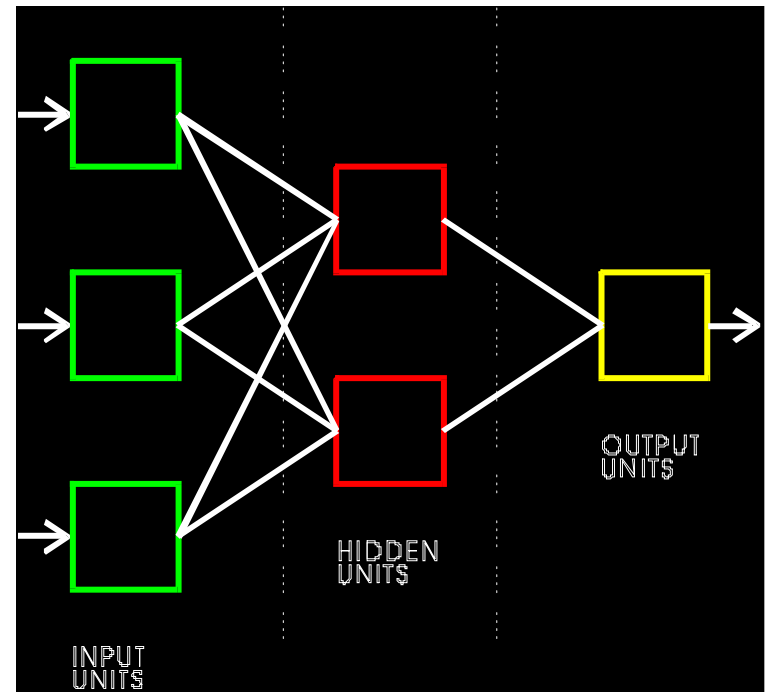
PSIPRED

- Works directly on PSI-BLAST profiles
- Uses 2-stages of feedforward neural networks
 - First network predicts secondary structure
 - Second network denoises outputs from 1st net
- Trained on ~1900 profiles
- Early stopped training method with 10% of training data kept aside to check convergence

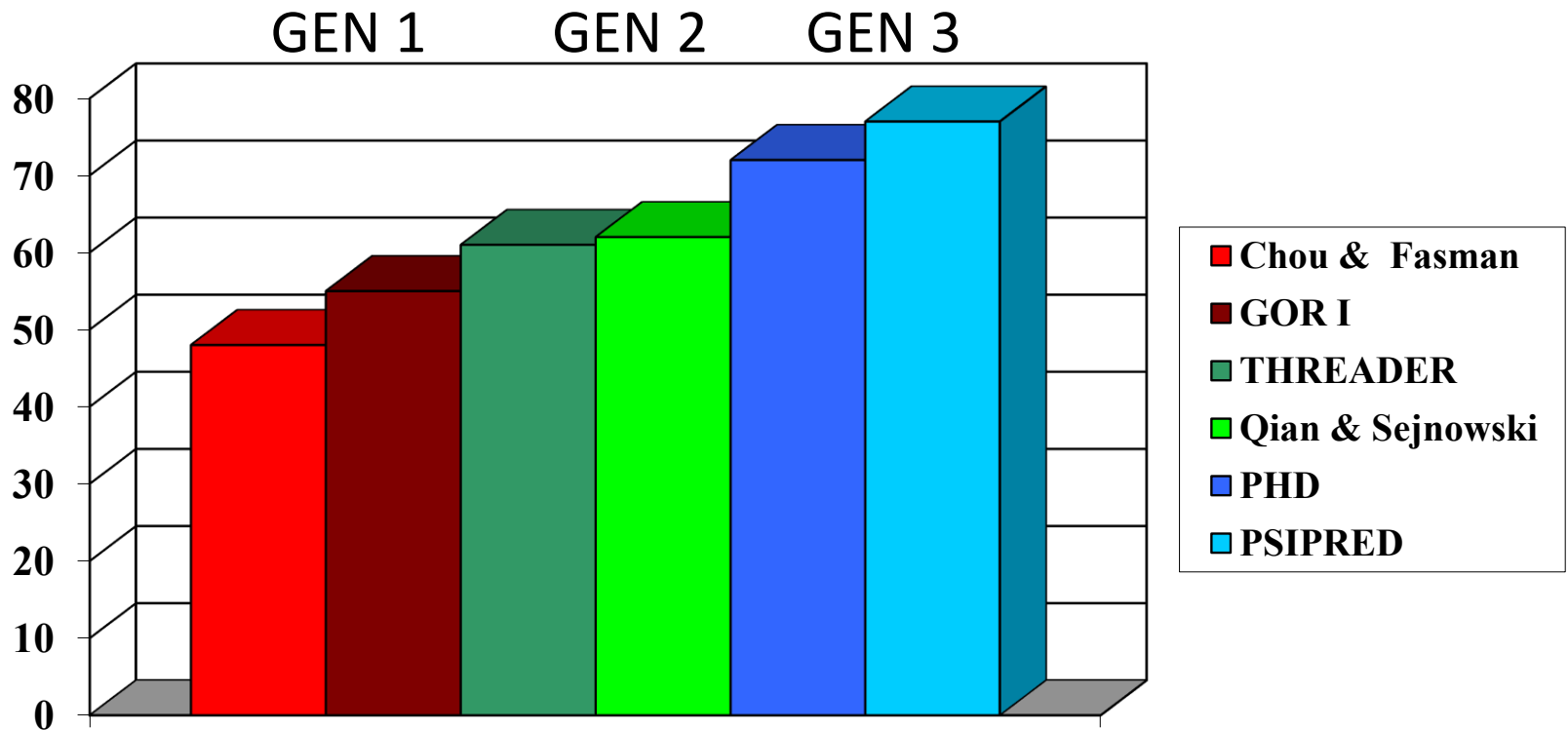
PSIPRED

Neural Networks

- 4 x 1st Networks
 - 15x21 Input units
 - 75 hidden units
 - 3 output units (H/E/C)
- 2nd Network
 - 15x4 Input units
 - 55 hidden units
 - 3 output units



Comparison of Generations by Average Q3 Scores



What about Deep Learning?

(Current state of the art)

- PSIPRED4 (2016)
 - 1-D convolutional layers plus fully connected layers
 - Achieves $\approx 83\%$ accuracy
- Deep(er) learning has not advanced the state of the art significantly beyond this
 - DeepCNF : Deep convolutional nets ($\approx 82\%$ accuracy)
 - SPIDER3 : Recurrent LSTM models ($\approx 84\%$ accuracy)
 - Results may well be overestimated due to flawed cross-validation
- Suggests either that all the local sequence information is already being exploited by shallower models or there is simply not enough data to effectively train deeper models

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Google's DeepMind predicts 3D shapes of proteins

AI program's understanding of proteins could usher in new era of medical progress



▲ Google's DeepMind artificial intelligence program, AlphaGo, plays South Korean professional Go player, Lee Sedol. Photograph: Alan Young-joon/AP

most viewed



Watson tells Corbyn he must change direction to stop Labour splitting



The first step for Labour's exiles: bring 29 more with them



Live Labour MPs applaud colleagues who quit at party meeting - as it happened



'Good riddance': Penistone locals have their say on Angela Smith resignation



Cabinet ministers tell May: stop using no-deal threat to negotiate

The New York Times

Making New Drugs With a Dose of Artificial Intelligence

T.M. Detwiler

By Cade Metz

Feb. 5, 2019

SAN FRANCISCO — You can think of it as a World Cup of biochemical research.

Every two years, hundreds of scientists enter a global competition. Tackling a biological puzzle they call “the protein folding problem,” they try to predict the three-dimensional shape of proteins in the human body. No one knows how to solve the problem. Even the winners only chip away at it. But a solution could streamline the way scientists create new medicines and fight disease.

Ian Sample Science editor

@iansample

Sun 2 Dec 2018 20:55 GMT

Having laid waste to the Atari classics and reached superhuman performance in chess and the Chinese board game, Go, Google's DeepMind outfit has turned its artificial intelligence on one of the toughest problems in science.

The result, perhaps, was predictable. At an international conference in Cancun on Sunday, organisers announced that DeepMind's latest AI program, AlphaFold, had beaten all-comers at a particularly fiendish task: predicting the 3D shapes of proteins, the fundamental molecules of life.

Science

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Complex of bacteria-infecting viral proteins modeled in CASP 13. The complex contains four separate subunits that were modeled individually. PROTEIN DATA BANK

Google's DeepMind aces protein folding

By Robert F. Service | Dec. 6, 2018, 12:05 PM

Turns out mastering chess and Go was just for starters. On 2 December, the Google-owned artificial intelligence firm DeepMind took top honors in the 13th Critical Assessment of Structure Prediction (CASP), a biannual competition aimed at predicting the 3D structure of proteins.

The contest worked like this: Competing teams were given the linear sequence of amino acids for 90 proteins for which the 3D shape is known but not yet published. Teams then computed how those sequences would fold. Though London-based DeepMind had not previously joined this competition, the predictions of its AlphaFold software were, on average, more accurate than those of its 97 competitors.

How close was the race? By one metric, not very. For protein sequences for which no other information was known—43 of the 90—AlphaFold made the most accurate prediction 25 times. That far outpaced the second place finisher, which won three of the 43 tests.

NEWS

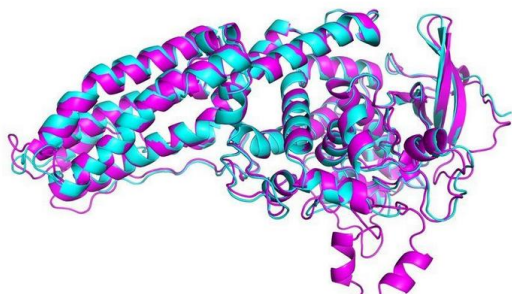
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Science & Environment

One of biology's biggest mysteries 'largely solved' by AI

By Helen Briggs
BBC science correspondent

30 November 2020



CASP/DEEPMIND/VTAGLIABRACCIDOMCHICK,UT SOUTHWESTE

A DeepMind model of a protein from the Legionnaire's disease bacteria (Casp-14)

nature > news > article

NEWS 30 NOVEMBER 2020

'It will change everything': DeepMind's AI makes gigantic leap in solving protein structures

Google's deep-learning program for determining the 3D shapes of proteins stands to transform biology, say scientists.

Ewen Callaway



A protein's function is determined by its 3D shape. Credit: DeepMind

An artificial intelligence (AI) network developed by Google AI offshoot DeepMind has made a gargantuan leap in solving one of biology's grandest challenges – determining a protein's 3D shape from its amino-acid sequence.

DeepMind's program, called AlphaFold, outperformed around 100 other teams in a biennial protein-structure prediction challenge called CASP, short for Critical Assessment of Structure Prediction. The results were announced on 30 November, at the start of the conference – held virtually this year – that takes stock of the exercise.

"This is a big deal," says John Moult, a computational biologist at the University of Maryland in College Park, who co-founded CASP in 1994 to improve computational methods for accurately predicting protein structures. "In some sense the problem is solved."

PDF version

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AI protein-folding algorithms solve structures faster than ever



The revolution will not be crystallized: a new method sweeps through structural biology



The computational protein designers



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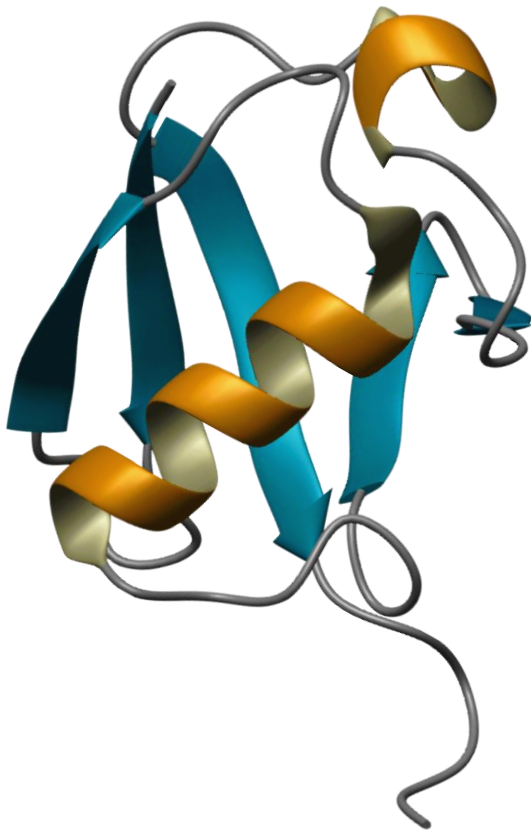
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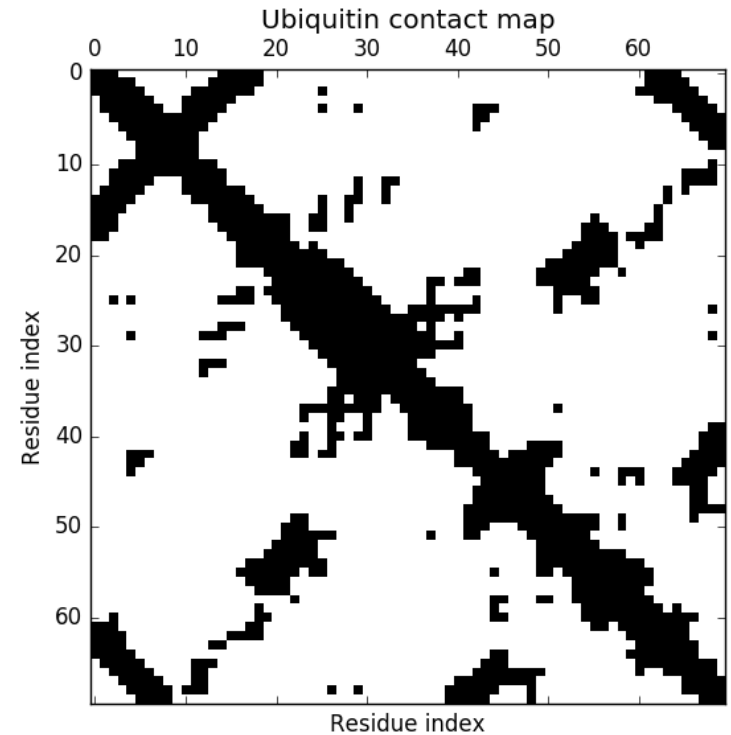
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Drug discovery

Protein Folding as a Graphical Modelling Problem

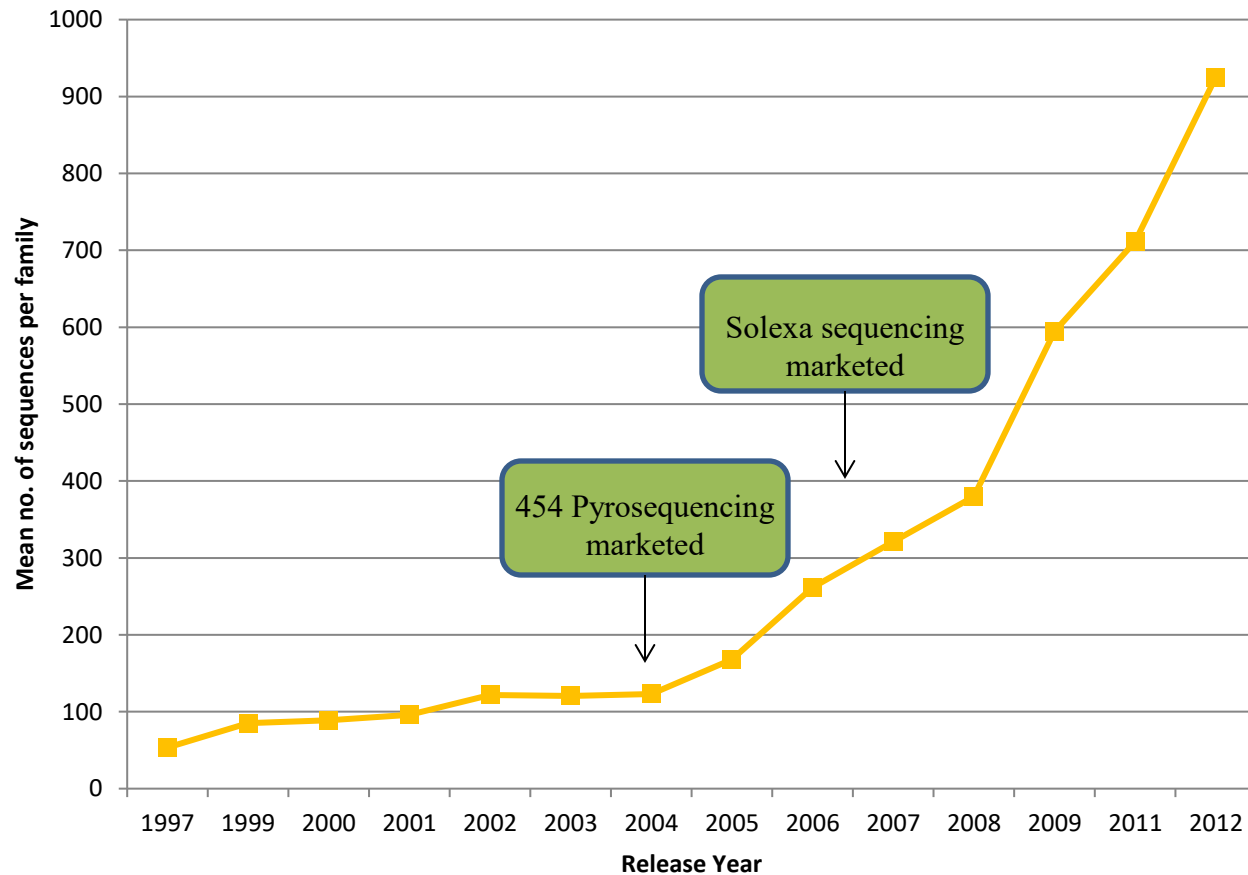


Calculate
distances from
coordinates
and apply a
threshold



The contact map can be
thought of as an adjacency
matrix.

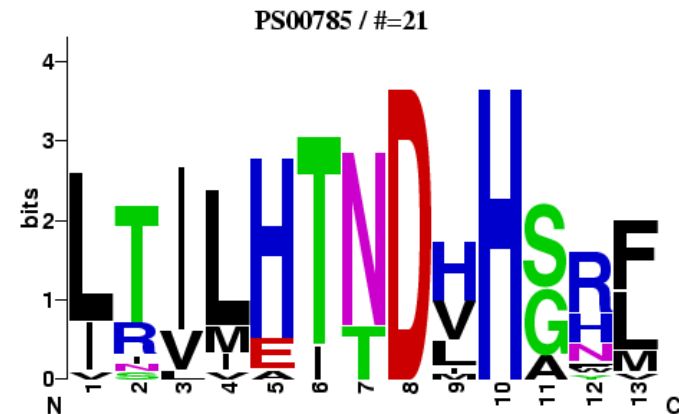
Growth of Sequence Family Size in Pfam



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16-----AELEWNNDDYVGVADIDEQHLELVALINQLHRSHKASKLDELAETRTTHFAIEERLMRESNYPDYESHRSYHEALIDQIRALQAKLDGGELHFLRTWLIRHICDVKQFG--
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19-----AFEPMEIHINEVKLLEDLLNENVKERFENFEDYKXHFAPFEEQMKKYNFFAYVPHKMEHDKIINQLNEVSKHLDDTLNETFTTWLINHIETIDTWT--
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25-----AIADWELKVGDIETIDAHKELIGLINDFEAKAKAPELERLQLYAYDHFAREEYIQAQAKYEGLEENKQHAALRTTLGTIYIEKFNAGQSAFLNHWLMNHILETDLKMRGK--
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Why having so many sequences is a game changer in bioinformatics

- In most molecular evolution applications a common simplifying assumption is made that mutations occurring at one site in a protein are independent of mutations occurring in other sites
- This simplification allows the use of Markovian methods e.g. HMMs and profiles
- This is also the best we can do with limited data
- With lots of sequence data, however, we can start considering coevolutionary (epistatic) effects



A lot of numbers

- A sequence profile or HMM for a protein of length 100 would have around $100 \times 21 = 2100$ parameters
- In the pairwise epistatic model we would have
 $\frac{1}{2} \times 100 \times 99 \times 21 \times 21 = 2,182,950$ parameters

That's why we need lots of sequence data!

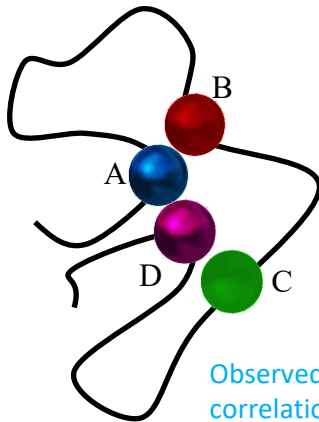
Correlated mutations in Proteins



- Residues in close proximity have a tendency to covary in order to maintain a microenvironment
- Changes at one site can be compensated for by a mutation in another position
- These structural constraints leave a record within the sequence
- By observing patterns of covarying residues in Multiple Sequence Alignments (MSAs) of homologous sequences, you can infer this structural information

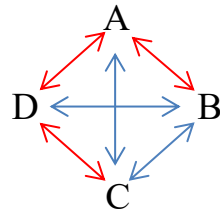
Contact Correlation Chains

Observed Residue Contacts



Observed residue mutations form a chain of correlations even though only 3 physical contacts are made out of the 6 possible pairs

Observed Correlations



Predicted Contacts

	A	B	C	D
A		Red	Blue	Red
B	Red		Blue	Blue
C	Blue	Blue		Red
D	Red	Blue	Red	

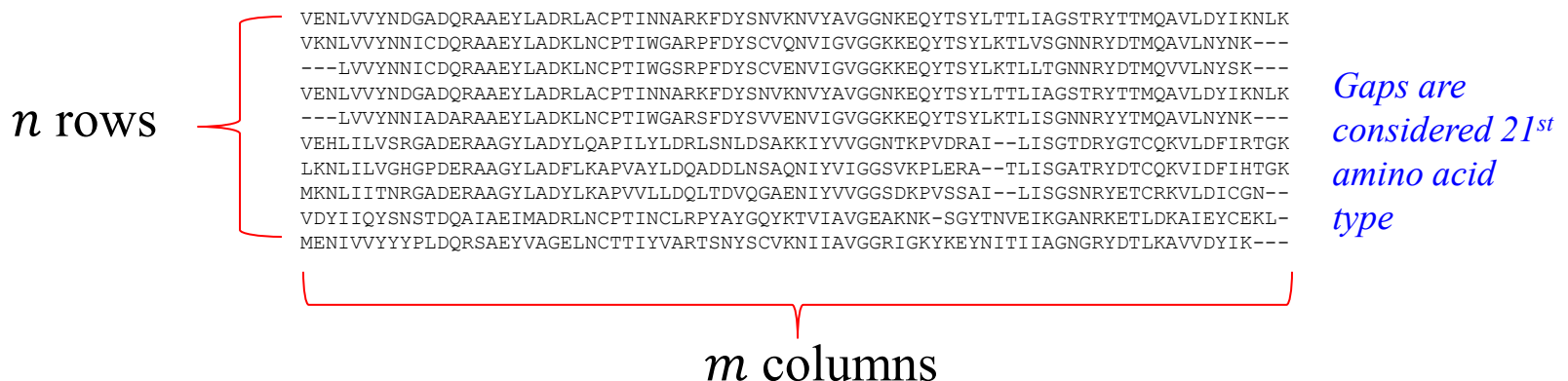
- Not all correlations are the result of direct interactions!
- Direct contact between A and B creates AB correlation
- Direct contact between A and D creates AD correlation
- Direct contact between C and D creates CD correlation
- But - this creates apparent correlation between AC, BC, and BD despite not being in contact
- Analogous to causative/transitive effects in gene network reconstruction

Calculation of Residue Covariation using a Sample Covariance Matrix

We consider an alignment with m columns and n rows, where each row represents a different homologous sequence and each column a set of equivalent amino acids across the evolutionary tree, with gaps considered as an additional amino acid type. We can compute a $21m$ by $21m$ sample covariance matrix as follows:

$$S_{ij}^{ab} = \frac{1}{n} \sum_{k=1}^n (x_i^{ak} - \bar{x}_i^a)(x_j^{bk} - \bar{x}_j^b)$$

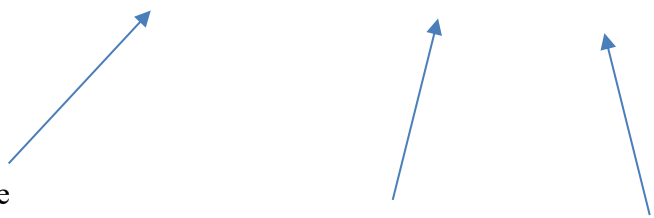
Where x_i^{ak} is a binary variable ($x \in \{0,1\}$) indicating the presence or absence of amino acid type a at column i in row k and x_j^{bk} the equivalent variable for observing residue type b at column j in row k .



Calculating the sample covariance matrix

- Consider an alignment with m columns and n rows, where each row represents a different homologous sequence and each column a set of equivalent amino acids across the evolutionary tree.
- We can compute a $21m$ by $21m$ sample covariance matrix as follows:

$$S_{ij}^{ab} = E(x_i^a x_j^b) - E(x_i^a)E(x_j^b) = f(A_i B_j) - f(A_i)f(B_j)$$



This is the probability of seeing residue type A at position i *and* residue type B at position j

This is the probability of seeing residue type A at position i

This is the probability of seeing residue type B at position j

Inverting the Covariance Matrix gives the Precision Matrix (Θ)

- Given the true covariance matrix, partial correlation coefficients (ρ_{ij}) can be calculated for all pairs of variables by calculating the matrix inverse of the covariance matrix.
- Each element in S^{-1} is zero if the partial covariance (correlation) of X_i and X_j with respect to the remaining variables is zero.
- For amino acid covariance matrices the partial correlation matrix gives the correlation between any pair of amino acids at any two sites, *conditional on the frequencies of amino acids at all other sites*.
- The precision matrix will be non-zero only for the direct correlations (contacts).

	A	B	C	D
A				
B				
C				
D				

$$\rho_{ij}=0$$

$$\rho_{ij}\neq 0$$

$$\Theta = S^{-1}$$

$$\rho_{ij} = - \frac{\Theta_{ij}}{\sqrt{\Theta_{ii}\Theta_{jj}}}$$

HOWEVER... the sample covariance matrix is often not invertible!

In this application it is always singular because not every amino acid is observed at every site.

Sparse Inverse Covariance Estimation (SICE)

avoids the need for matrix inversion and
overcomes the problem of over-fitting

- Even if we find a way of inverting the covariance matrix, it will suffer badly from **overfitting to the noise**
- Sparse Inverse Covariance Estimation allows us to avoid both these issues
- Has been used successfully in gene network reconstruction
- We use the **dual optimization formulation of Banerjee**, solved using the coordinate descent method of Friedman et al. (the graphical Lasso)

Banerjee, O., El Ghaoui, L. & d'Aspremont, A. (2008). Model selection through sparse maximum likelihood estimation. *J. Machine Learning Research*, **9**, 485-516.

Friedman, J., Hastie, T. & Tibshirani, R. (2008). Sparse inverse covariance estimation with the **graphical Lasso**. *Biostatistics*, **9**, 432-441.

SICE allows us to overcome the problem of finding the full matrix inverse

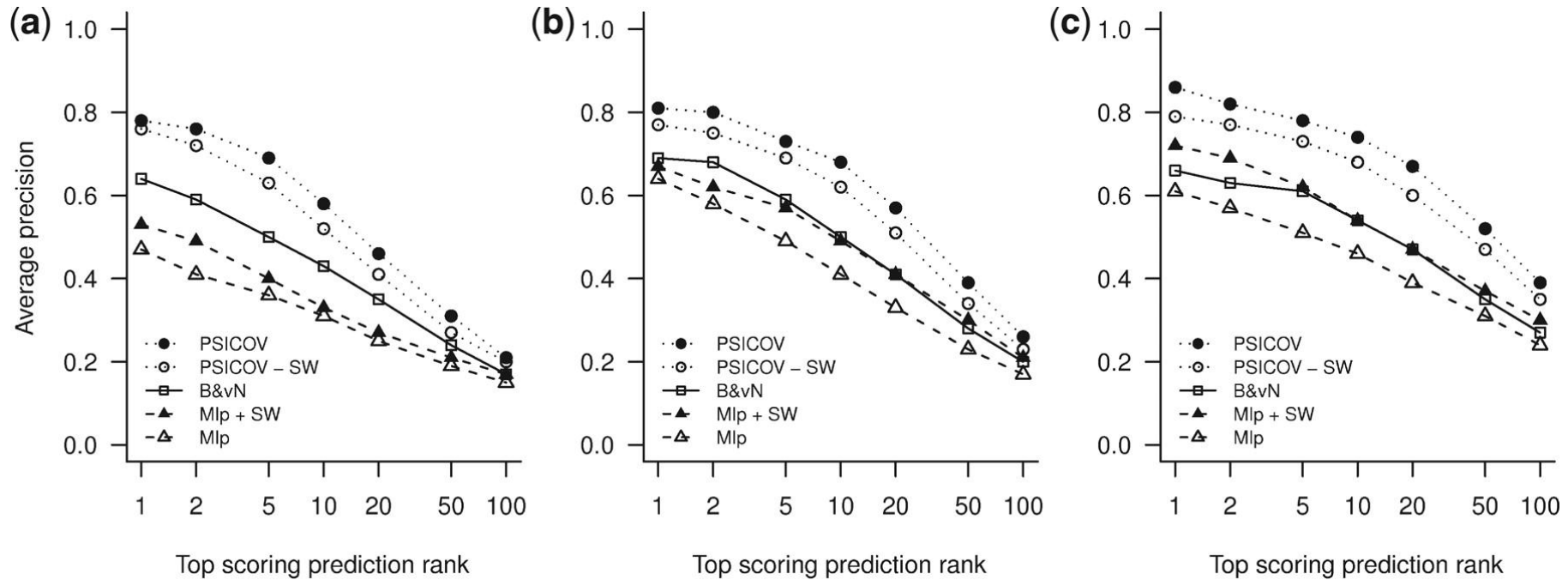
- The Graphical Lasso attempts to find a sparse inverse covariance matrix by minimization of a specific objective function.
- **The expected sparsity of the inverse covariance matrix itself provides a powerful self-contained constraint on the obtained solution and thus helps avoid over-fitting.**
- In general terms, where an inverse covariance estimate is constrained to be sparse, the non-zero terms tend to more accurately relate to correct positive correlations in the true inverse covariance matrix. SICE therefore not only solves the inversion problem but also gives better answers!
- The assumption of sparsity in this application is well justified from observations of contacts in known protein structures, where on average only ~3% of all residue pairs are observed to be in direct contact. We actually use this observation to tune the regularisation parameter for each case.

Let S be the empirical covariance matrix computed from a sequence of d -dimensional vectors, $x^1, \dots, x^n$, sampled from some fixed but unknown probability distribution. The graphical Lasso estimates the inverse covariance of the data by minimising the objective function:

$$\sum_{ij=1}^d S_{ij}\theta_{ij} - \log \det \Theta + \rho \sum_{ij=1}^d |\theta_{ij}|$$

This term is the ℓ_1 -norm of matrix Θ and relates to the sparsity of the matrix

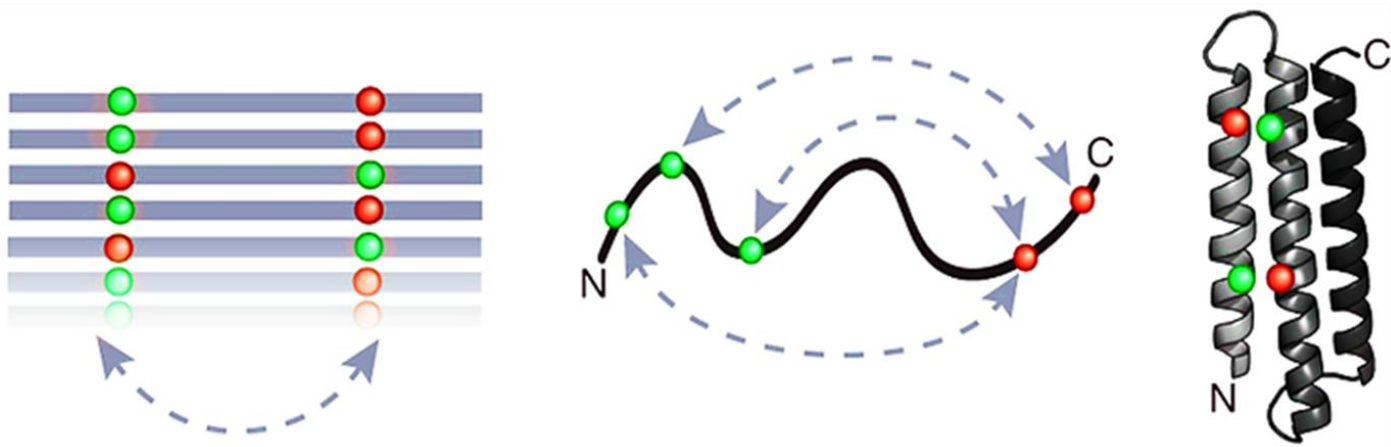
PSICOV contact prediction performance



- The three panels indicate the average precision of the top n ranked predictions at sequence separations between 5 and 9 residues (a), between 10 and 23 residues (b) and >23 residues (c).
- In some cases, accuracy of long-range contacts can approach 80% - an improvement sufficient to be of significant benefit in protein structure prediction.

Predicting the 3-D structure of proteins by co-evolution

- We can produce accurate lists of contacting residues from covariation observed in **large multiple sequence alignments**
- If we have an efficient way to project this information into 3-D space whilst satisfying the **physicochemical constraints of protein chains** then we have everything we need to predict 3-D structure!



Deep learning allows us to predict inter-residue *distances* rather than just contacts

- By combining deep learning and coevolution we can now predict the likely distances between amino acids rather than just binary contacts

DMPfold (open source)

**nature
communications**

Article | [Open Access](#) | Published: 04 September 2019

Deep learning extends de novo protein modelling coverage of genomes using iteratively predicted structural constraints

Joe G. Greener, Shaun M. Kandathil & David T. Jones 

Nature Communications **10**, Article number: 3977 (2019) | [Cite this article](#)

AlphaFold (proprietary)

nature

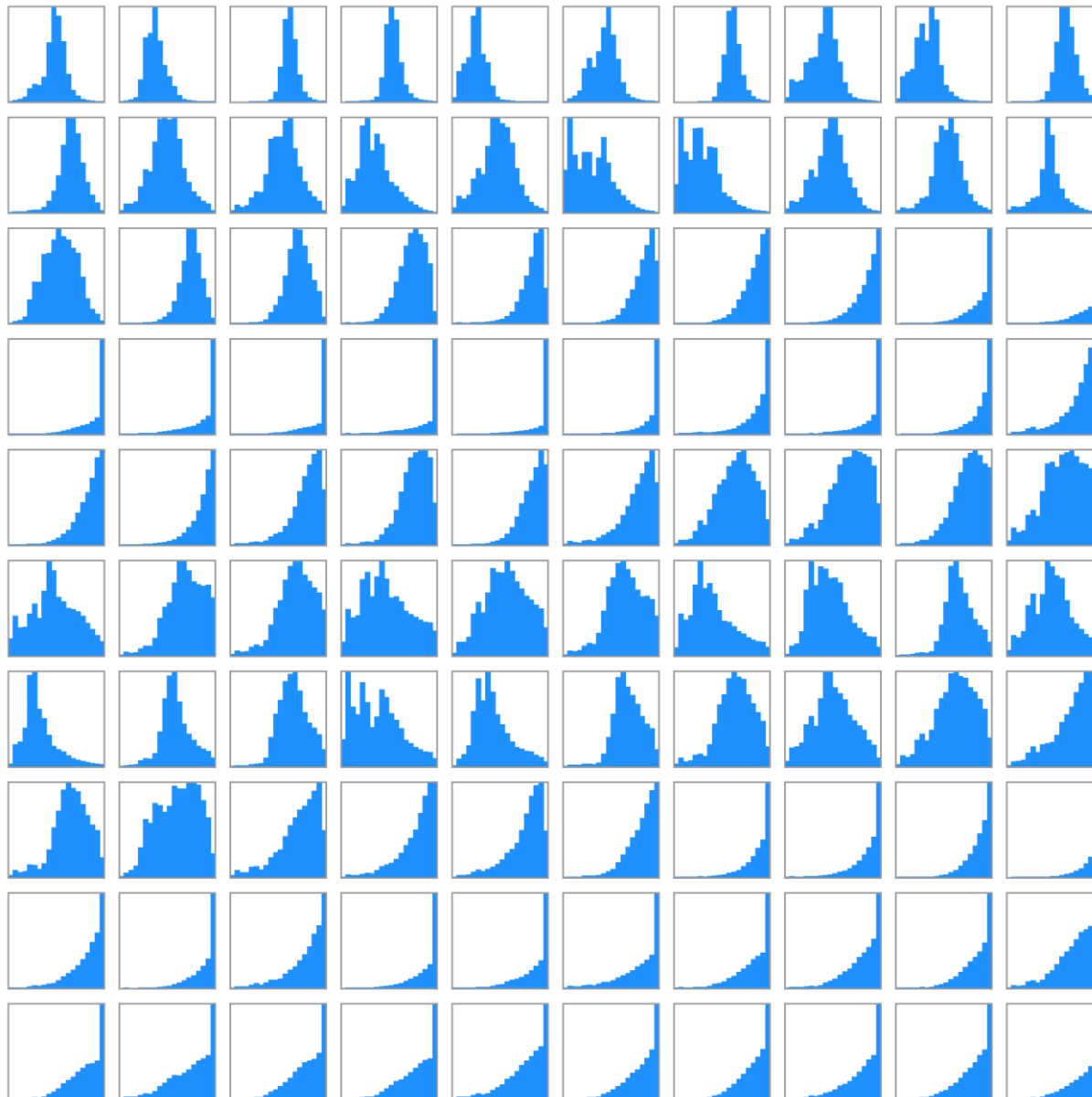
Article | Published: 15 January 2020

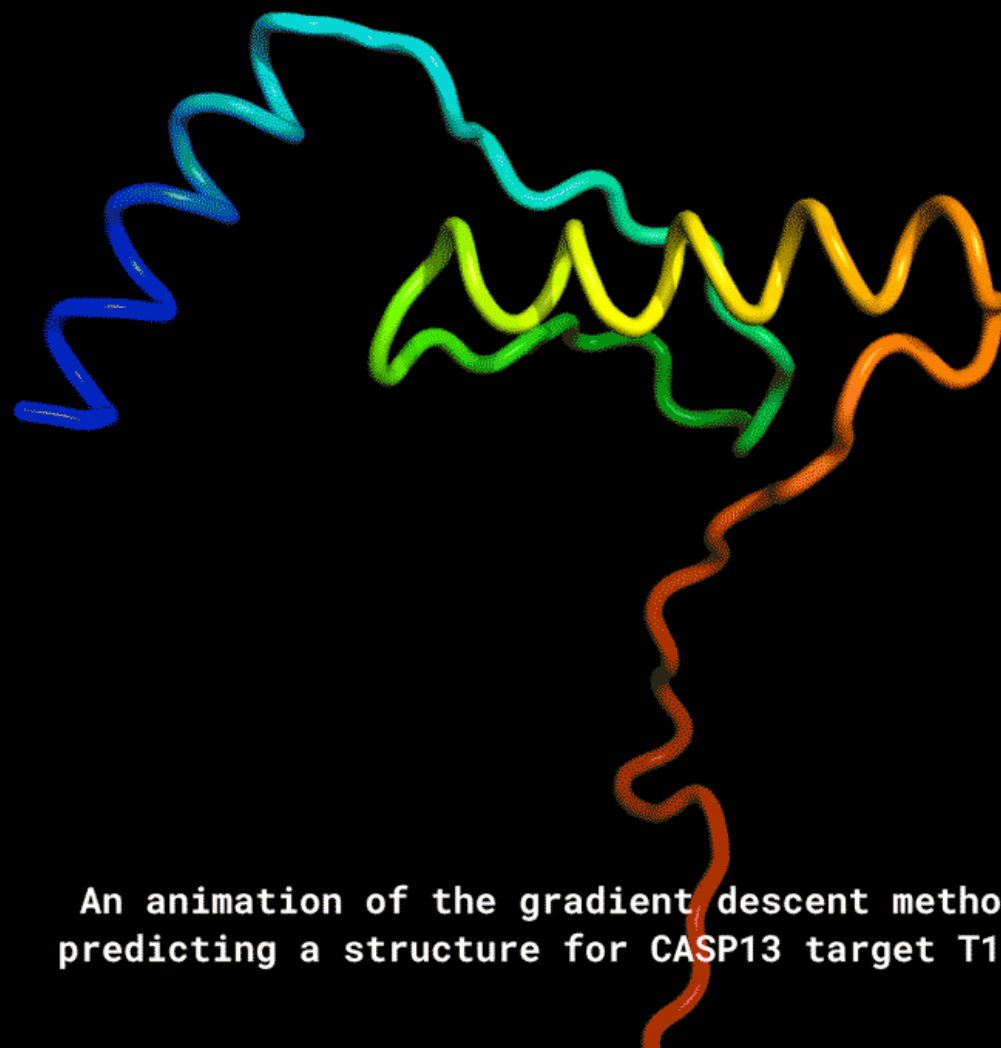
Improved protein structure prediction using potentials from deep learning

Andrew W. Senior , Richard Evans, John Jumper, James Kirkpatrick, Laurent Sifre, Tim Green, Chongli Qin, Augustin Židek, Alexander W. R. Nelson, Alex Bridgland, Hugo Penedones, Stig Petersen, Karen Simonyan, Steve Crossan, Pushmeet Kohli, David T. Jones, David Silver, Koray Kavukcuoglu & Demis Hassabis

Nature **577**, 706–710(2020) | [Cite this article](#)

Example Predicted Distance Distributions from Pfam Family PF04409





An animation of the gradient descent method
predicting a structure for CASP13 target T1008

NEWS

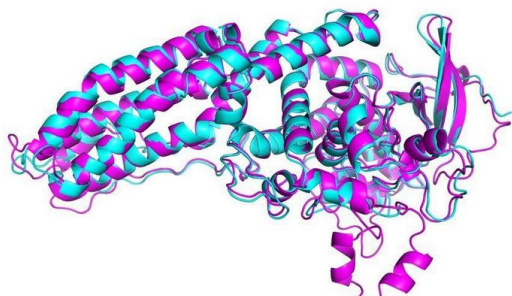
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Science & Environment

One of biology's biggest mysteries 'largely solved' by AI

By Helen Briggs
BBC science correspondent

30 November 2020



CASP/DEEPMIND/VTAGLIABRACCIDOMCHICK,UT SOUTHWESTE

A DeepMind model of a protein from the Legionnaire's disease bacteria (Casp-14)

nature > news > article

NEWS 30 NOVEMBER 2020

'It will change everything': DeepMind's AI makes gigantic leap in solving protein structures

Google's deep-learning program for determining the 3D shapes of proteins stands to transform biology, say scientists.

Ewen Callaway



A protein's function is determined by its 3D shape. Credit: DeepMind

An artificial intelligence (AI) network developed by Google AI offshoot DeepMind has made a gargantuan leap in solving one of biology's grandest challenges – determining a protein's 3D shape from its amino-acid sequence.

DeepMind's program, called AlphaFold, outperformed around 100 other teams in a biennial protein-structure prediction challenge called CASP, short for Critical Assessment of Structure Prediction. The results were announced on 30 November, at the start of the conference – held virtually this year – that takes stock of the exercise.

"This is a big deal," says John Moult, a computational biologist at the University of Maryland in College Park, who co-founded CASP in 1994 to improve computational methods for accurately predicting protein structures. "In some sense the problem is solved."

PDF version

RELATED ARTICLES

AI protein-folding algorithms solve structures faster than ever



The revolution will not be crystallized: a new method sweeps through structural biology



The computational protein designers



Revolutionary microscopy technique sees individual atoms for first time



SUBJECTS

Computational biology and bioinformatics

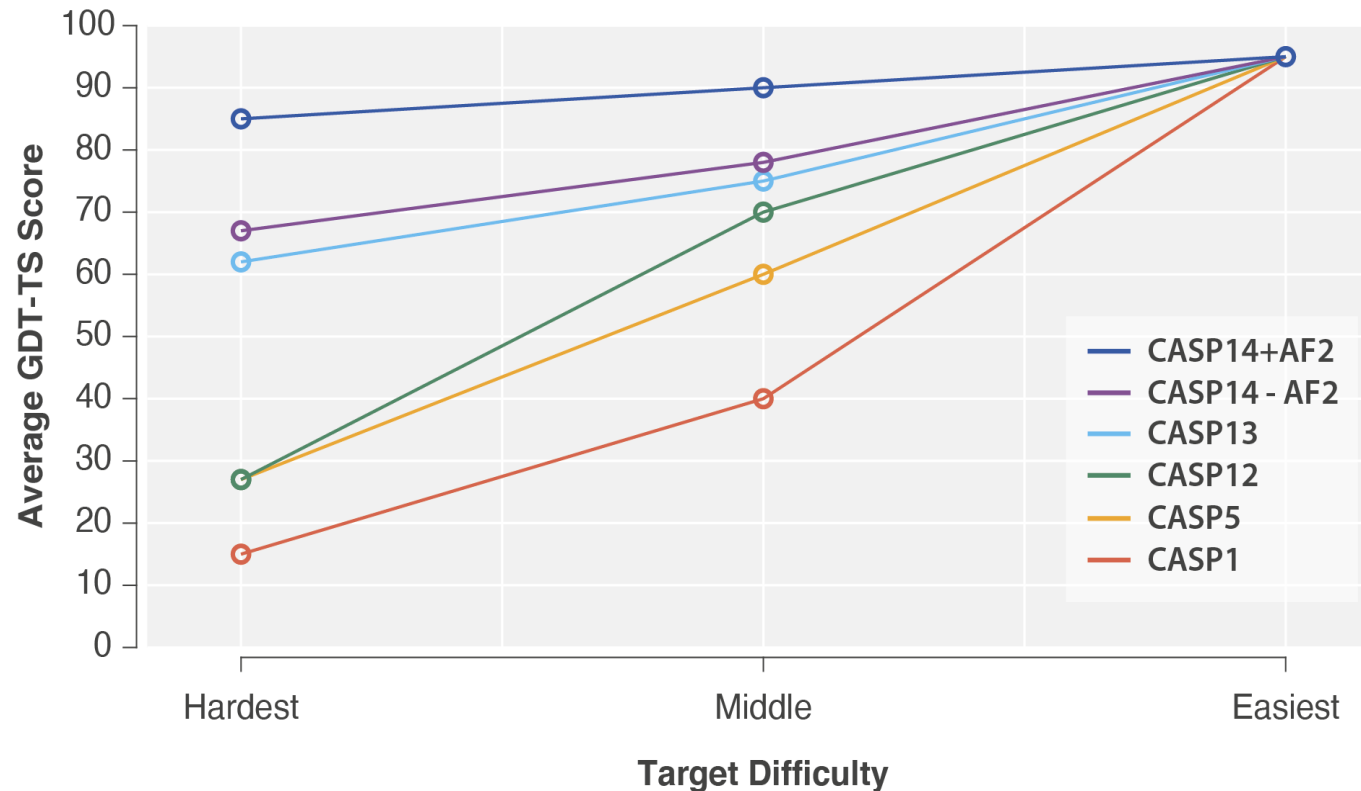
Structural biology

Drug discovery

Differences between AlphaFold1 and AlphaFold2

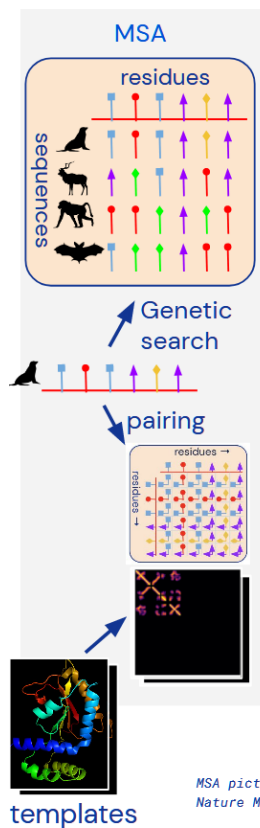
- AlphaFold2 implements an *end-to-end* model
 - No preprocessing of inputs (apart from MSA generation)
 - MSAs are input directly to the model
 - No external structure generating step is required
- Architecture is based on self-attention rather than ResNets

The impact of AlphaFold2 on CASP

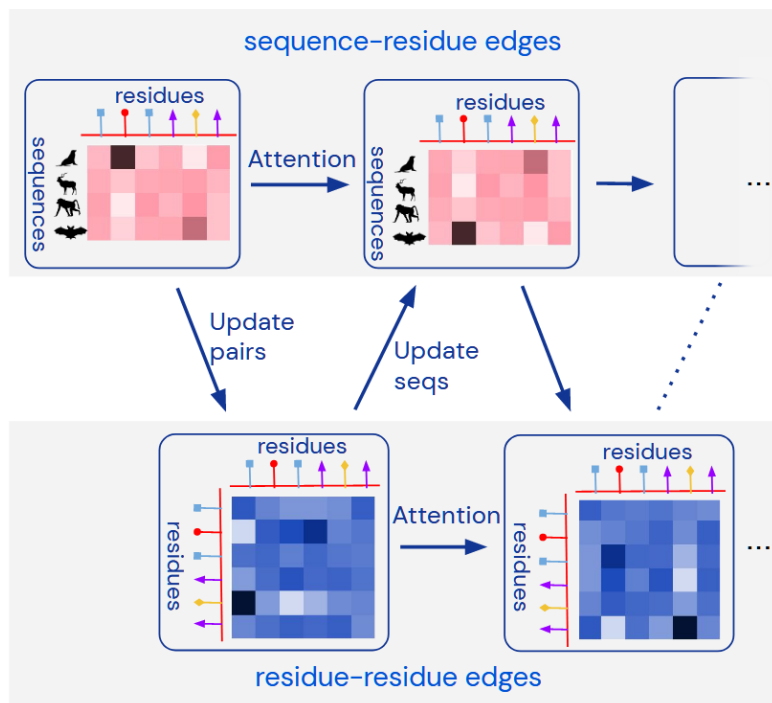


AlphaFold2 in Outline

Embedding

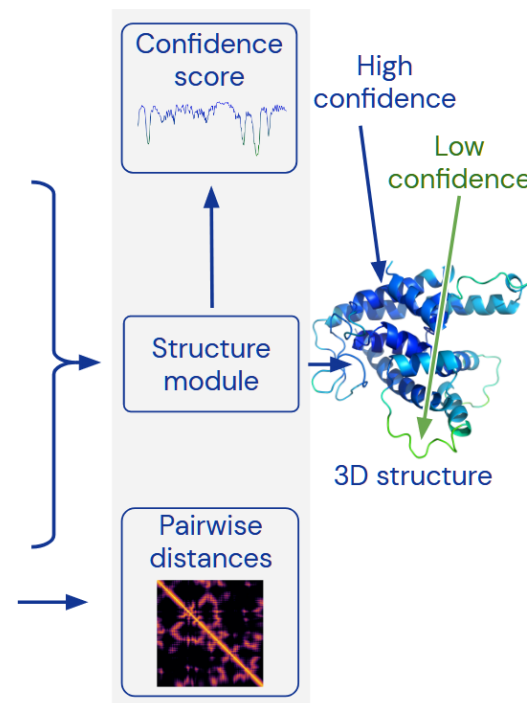


Trunk



Heads

© 2020 DeepMind Technologies Limited

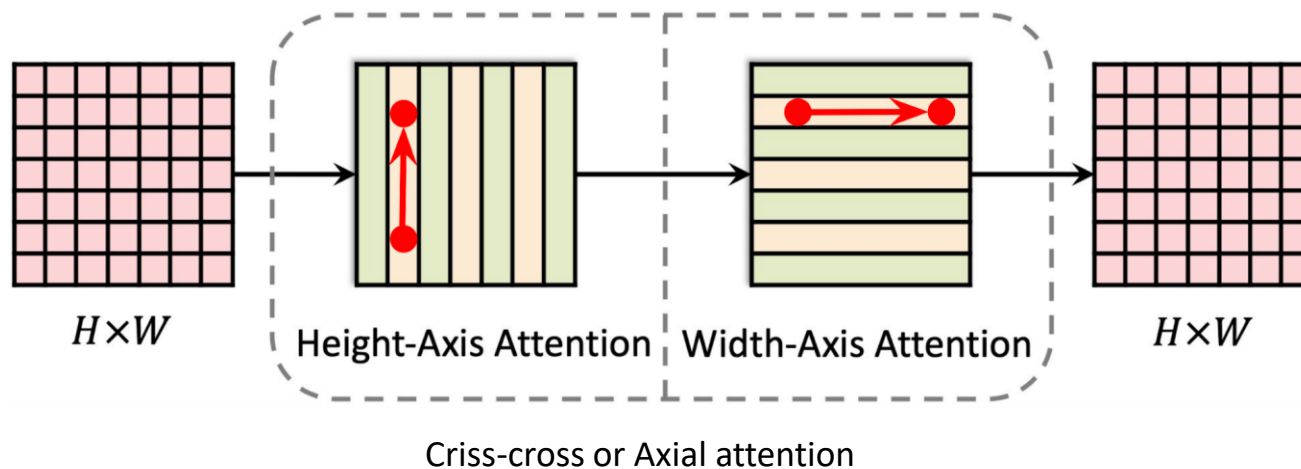


MSA picture inspired by: RieseIman, A.J., Ingraham, J.B. & Marks, D.S.,
Nature Methods (2018) doi:10.1038/s41592-018-0138-4

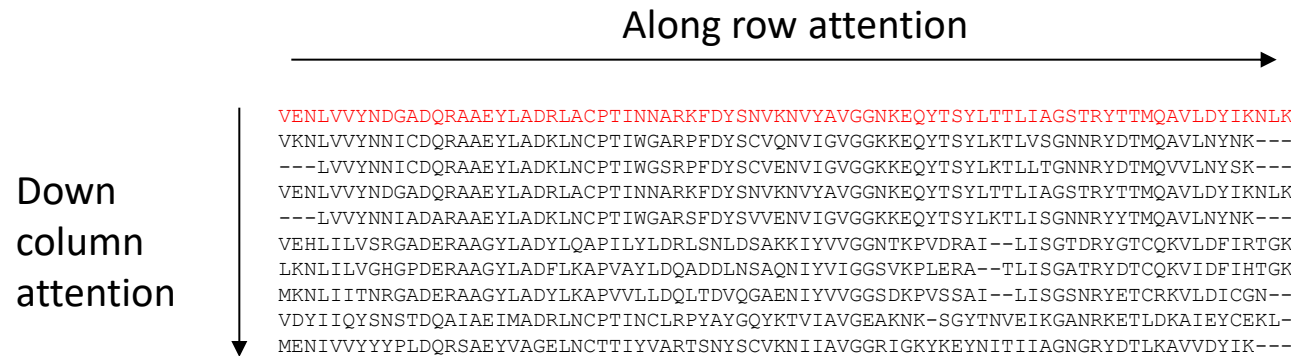


Attention in more than one dimension

- Applying attention to more than one dimension e.g. for images can be done by running separate transformers along each axis
- Originally applied to images i.e. 2-D



Transformers for MSAs



TARGET
SEQUENCE

MSA Transformer

Roshan Rao^{1,2} Jason Liu³ Robert Verkuil³ Joshua Meier³
John F. Canny¹ Pieter Abbeel¹ Tom Sercu³ Alexander Rives^{3,4}

Abstract

Unsupervised protein language models trained across millions of diverse sequences learn structure and function of proteins. Protein language models studied to date have been trained to perform inference from individual sequences. The longstanding approach in computational biology has been to make inferences from a family of evolutionarily related sequences by fitting a model to each family independently. In this work we combine the two paradigms. We introduce a pro-

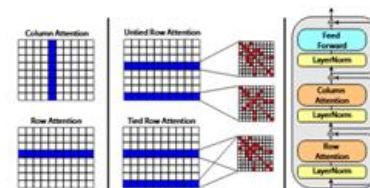


Figure 1. Left: Sparsity structure of the attention. By constraining attention to operate over rows and columns, computational cost

Technical
note:

Columns with
gaps in the
target
sequence are
often
removed for
simplicity

BERT can again be used for training...

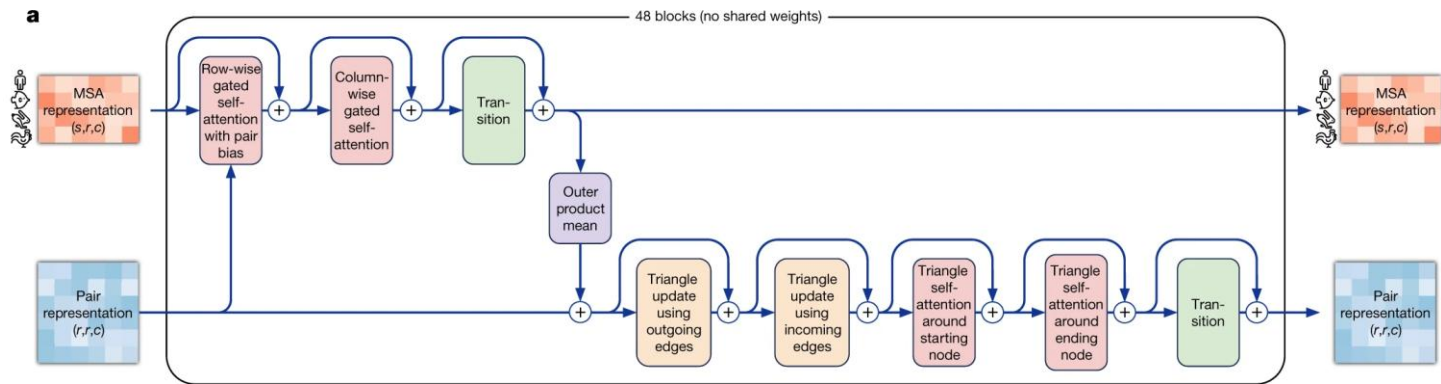
VENLVVYNDGADQRA#EYLADRLACPTINNARKFDYSNVKNVYAVGGNKEQYTSYLTTLIAGSTRYTTMQAVLDYIKNLK
VKNLVVYNNICDQRAAEYLADKLNC#TIWGARPFDYSCVQNVIGVGGKKEQYTSYLKTLVSGNNRYDTMQA#LNYNK---
---LVVYNNICDQRAAEYLADKLNCPTIWGSRPFDYSCVENVIGVGGK#EQYTSYLKTLTGNRYDTMQVVLNYSK---
VENLVVYNDGADQ#AAEYLADRLACPTINNARKFDYSNVKNVYAVGGNKEQYTSYLTTLIAGSTRYTTMQAVLDYIKNLK
---LVVYNNIADARAAEYLADKLNCPTIWGARSFDYSVVENVIGVGGKKEQYTSYLKTLISGNNRYYTMQAVLNYNK---
VEHLILVSRGADERAAGYLADYLQAPILYLDRLSNLDSAKKIYVVGGNTPVDRAI--LISGTDRYGTCQKVLDFIHTGK
LKNLIL#GHGPDERAAGYLADFLKAPVAYLDQADDLNSAQNIYVIGGSVKPLERA--TLISGAT#YDTCQKVIDFIHTGK
MKNLIITNRGADERAAGYLADYLKAPVVLLDQLTDVQGAENIYVVGGS DKPVSSAI--LISGSNRYETCRKVLDICGN--
VDYIIQYSNSTDQAI AEIMADRLNCPTINCLRPYAYGQYKT VIAVGEAKNK-SGYTNVEIKGANRKETLDKAIEYCEKL-
MENIVVYYYPLDQRS AEYVAGELNCTTIYVAR#SNYSCVKNI IAVGGRIGKYKEYNITIIAGNGRYDTLKA#VDYIK---

= Masked

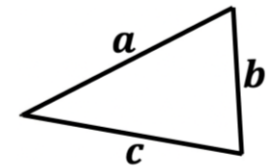
BERT can again be used for training...

VENLVVYNDGADQRAAEYLADRLACPTINNARKFDYSNVKNVYAVGGNKEQYTSYLTTLIAGSTRYTTMQAVLDYIKNLK
VKNLVVYNNICDQRAAEYLADKLNCP TIWGARPFDYSCVQNVIGVGGKKEQYTSYLKTLVSGNNRYDTMQAVLNYNK---
---LVVYNNICDQRAAEYLADKLNCP TIWGSRPFDYSCVENVIGVGGKKEQYTSYLKTLTGNRYDTMQVVLNYSK---
VENLVVYNDGADQRAAEYLADRLACPTINNARKFDYSNVKNVYAVGGNKEQYTSYLTTLIAGSTRYTTMQAVLDYIKNLK
---LVVYNNIADARAAEYLADKLNCP TIWGARSFDYSVVENVIGVGGKKEQYTSYLKTLISGNNRYYTMQAVLNYNK---
VEHLILVSRGADERAAGYLADYLQAPILYLDRLSNLDSAKKIYVVGGN TKPVDRAI--LISGTD RYGTCQKVLD FIRTGK
LKNLILVGHGPDERAAGYLADFLKAPVAYLDQADDLNSAQNIYVIGGSVKPLERA--TLISGATRYDTCQKVIDFIHTGK
MKNLIITNRGADERAAGYLADYLKAPVVLLDQLTDVQGAENIYVVGGS DKPVSSAI--LISGSNRYETCRKVL DICGN--
VDYIIQYSNSTDQAI AEIMADRLNCP TINCLRPYAYGQYKT VIAVGEAKNK-SGYTNVEIKGANRKETLDKAIEYCEKL-
MENIVVYYYPLDQRSAEYVAGELNCTTIYVARTSNYSCVKNI IAVGGRIGKYKEYNITIIAGNGRYDTLKAVVDYIK---

The AlphaFold EvoFormer Block



Main job of the pair track is just to keep track of the pairwise distances (edges in the structure graph) and also to provide positional encoding. Triangle self-attention attempts to encourage satisfaction of triangle inequalities for edges between all *triplets* of nodes.



$$\begin{aligned}a + b &> c \\a + c &> b \\b + c &> a\end{aligned}$$

AlphaFold2

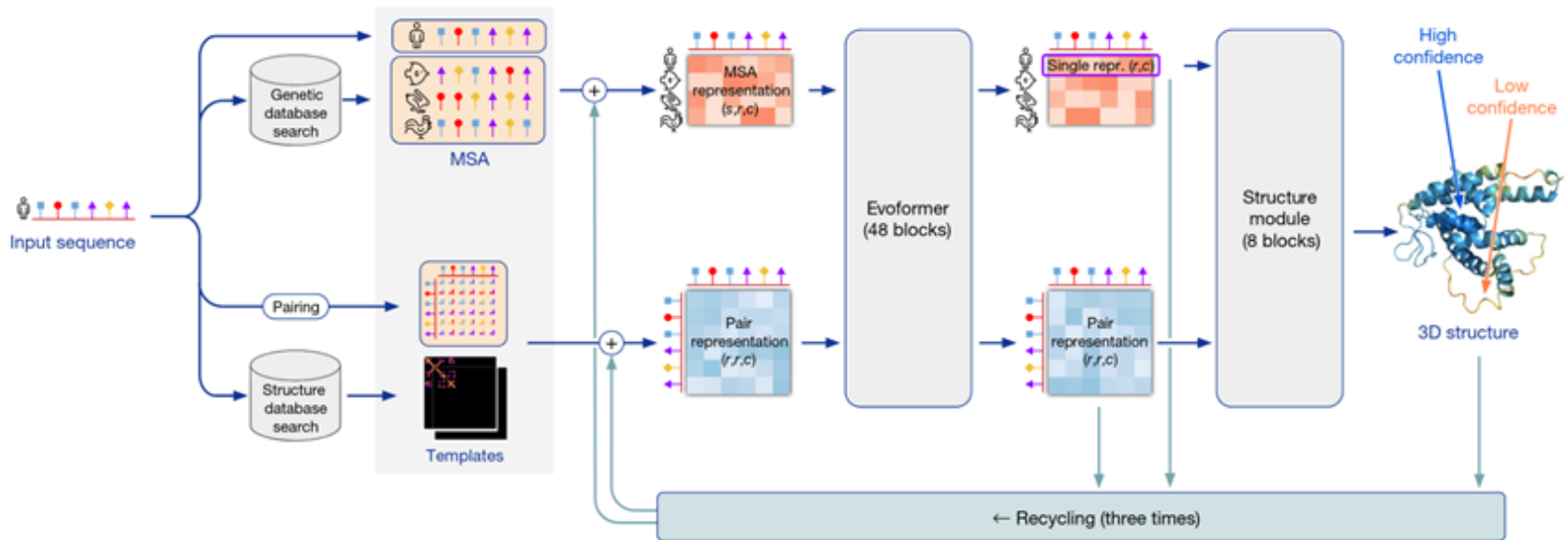
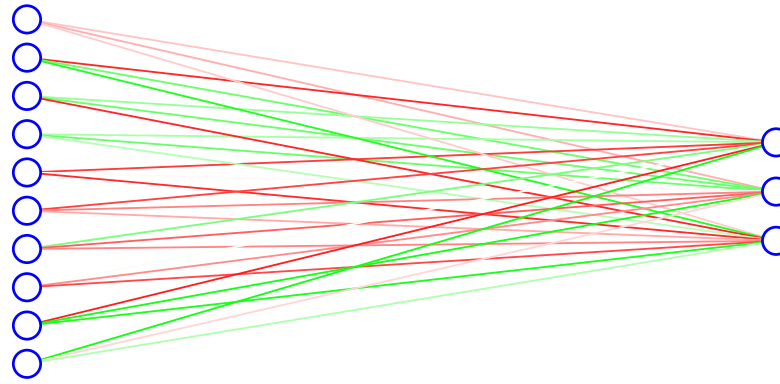


Diagram from: Jumper, J., et al., 2021, Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873), pp.583-589.

How do you produce a 3-D structure from all that?



This simple linear neural network reduces the number of input dimensions (10) down to 3.

Note how the three outputs are just weighted sums of the inputs. This operation is called a projection, and is used many times in AlphaFold2. AlphaFold2 internal dimensions are usually 256, but sometimes 384 or 128.

Projections can either reduce dimensions (like above) or increase them.

NOTE similarity to Principal Component Analysis (PCA) methods.

The Structure Module

- The output module of AlphaFold projects the embeddings to $C\alpha$ coordinates plus a rotation matrix
- The rotation matrix is used to rotate a rigid body comprising the N, $C\alpha$ and C atoms of the protein backbone
- It also predicts the side chain torsion angles (Chi angles)
- Form of SE(3) equivariant transformer model

→ **End-to-end folding** instead of gradient descent

→ **Protein backbone = gas of 3-D rigid bodies**
(chain is learned!)

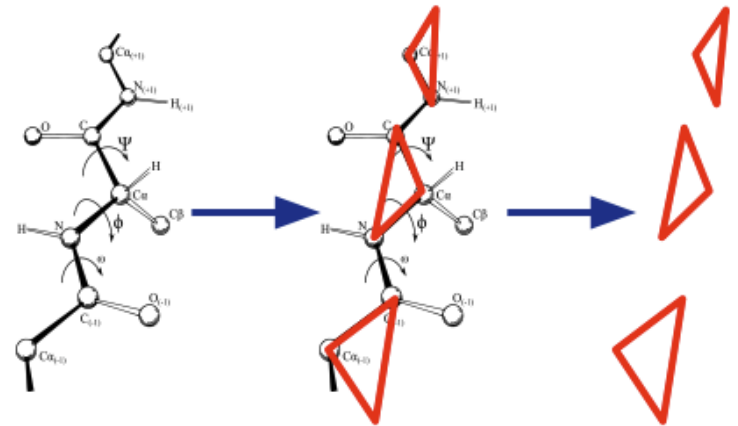


Image: Dcrjsr, vectorised Adam Rędzikowski (CC BY 3.0, Wikipedia)

Training AlphaFold2

- You need known 3-D structures
 - ~170,000 chains taken from PDB
- You need MSAs
 - HHblits to generate MSA for each chain in the training set
- Train AlphaFold to translate from a given input MSA to the correct native 3-D coordinates of the protein chain
- That's it!

Limitations of AlphaFold2

- Similar problems to older methods which rely on evolutionary information
- Model quality depends on having a good multiple sequence alignment
 - Some sequences have few homologs e.g. viral proteins
- Reliance on evolutionary information means that AF2 cannot predict mutation effects or things like antibody structures
- Only produces a single “maximum likelihood” conformation
 - Sometimes produces an average conformation i.e. one between two distinct conformations observed in PDB
 - Need better methods which can model conformational change

AlphaFold-Multimer can model multimers (quaternary structure)

- Co-evolution can be observed between protein chains that are in contact in multimers
- AlphaFold can model multimeric structures with properly paired MSAs
- Success rate is about 50%
 - Probably limited by MSA quality and size of interface regions
- Doesn't work for antibody-antigen docking

You can try AlphaFold yourself...

<https://github.com/sokrypton/ColabFold>